A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Kinetics

At AS you explained qualitatively why concentration, temperature and catalysts change the rate of a reaction. Here the treatment becomes quantitative: rate equations say exactly how the rate depends on each concentration, the orders they contain reveal which species take part in the slowest step of the mechanism, and the Arrhenius equation puts a number on the activation energy.

Rate of reaction

DEFINITION

The **rate of reaction** is the change in concentration of a reactant or product per unit time. Its usual unit is **mol dm^{-3} s^{-1}**.

On a graph of concentration against time, the rate at any moment is the **gradient** of the curve at that moment. Because the curve bends, you draw a **tangent** to it and find the tangent's gradient. A reactant's concentration falls, so its gradient is negative; the rate is the size of that gradient. The **initial rate** is the gradient of the tangent at t = 0, where the reaction is fastest because the reactants are most concentrated.

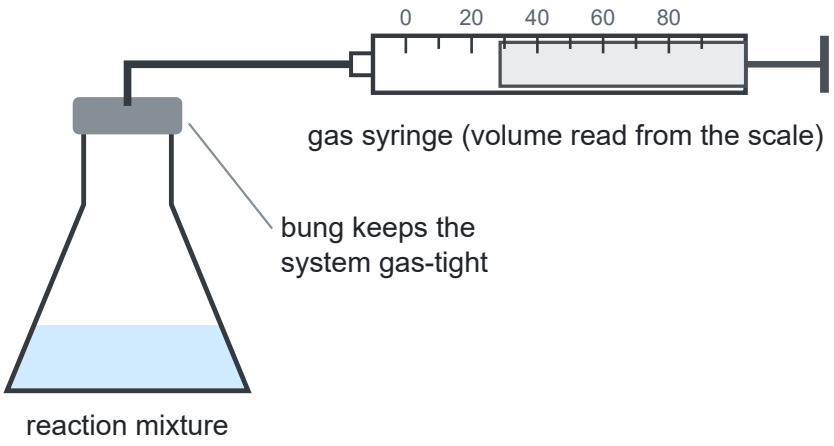
The rate in mol dm^{-3} s^{-1} depends on which substance you follow. In $2N_2O_5 \rightarrow 4NO_2 + O_2$, nitrogen dioxide forms twice as fast as N_2O_5 is used up and four times as fast as oxygen forms, so always say which substance a rate refers to.

Following the rate of a reaction

To follow a reaction you need a property that changes as it goes and that you can measure without disturbing it (or a way of stopping samples so you can analyse them). Choose the method to suit the reaction.

Method	Use it when…	Example
Volume of gas, collected in a gas syringe	a gas is given off	$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$
Loss of mass, on a balance	a gas escapes from an open flask; works best for a dense gas such as CO_2	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

Colorimetry	a reactant or product is coloured; absorbance is proportional to its concentration	$HCOOH(aq) + Br_2(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$: the orange bromine fades
Titration of quenched samples	an acid, alkali or iodine is used up or formed	$CH_3COCH_3 + I_2 \rightarrow CH_3COCH_2I + H^+ + I^-$: titrate the iodine left with thiosulfate
Electrical conductivity	the number, charge or type of ions changes	$(CH_3)_3CBr + H_2O \rightarrow (CH_3)_3COH + H^+ + Br^-$: ions are produced
"Clock" reaction	you only need the initial rate: time how long a fixed small amount of product takes to form	the iodine clock described in the core practical below



Collecting a gas in a gas syringe. Readings of volume against time give a curve whose gradient is the rate of gas production.

EXAM TIP

Quenching means stopping the reaction in a sample so its composition stays fixed while you analyse it. You can quench by diluting the sample in cold water, or by removing a catalyst: adding sodium hydrogencarbonate neutralises an acid catalyst.

Rate equations and orders

For a reaction between A and B, experiments show that the rate is proportional to each concentration raised to some power:

The rate equation

$$\text{rate} = k[\text{A}]^m[\text{B}]^n$$

$[\text{A}]$: concentration of A / mol dm^{-3} m, n: orders with respect to A and B k: rate constant

- **Order with respect to a reactant:** the power its concentration is raised to in the rate equation. For this course the order is 0, 1 or 2.
- **Overall order:** the sum of the individual orders, m + n.
- **Rate constant, k:** the constant of proportionality. It does not change with concentration, but it is different at each temperature, increasing as the temperature rises. A larger k means a faster reaction.

Order with respect to A	Rate is…	Double [A]	Triple [A]
0	independent of [A] (because $[A]^0 = 1$)	rate × 1	rate × 1
1	proportional to [A]	rate × 2	rate × 3
2	proportional to $[A]^2$	rate × 4	rate × 9

COMMON MISTAKE

Orders can only be found by **experiment**. They are not the numbers in front of the formulae in the balanced equation. For $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ the rate equation is rate = $k[H_2O_2][I^-]$: first order in iodide despite the 2, and zero order in H^+.

Units of the rate constant

The units of k depend on the overall order, so work them out from the rate equation each time.

Units of k when rate = k[A][B]²

1. Rearrange the rate equation to make k the subject:

$$k = \frac{\text{rate}}{[\text{A}][\text{B}]^2}$$

2 Substitute the units of each quantity:

$$k = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{(\text{mol}\,\text{dm}^{-3})(\text{mol}\,\text{dm}^{-3})^2} = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{\text{mol}^3\,\text{dm}^{-9}}$$

3 Cancel one mol dm^{-3} from top and bottom, then move what is left on the bottom to the top by changing the signs of its powers:

$$k = \frac{\text{s}^{-1}}{\text{mol}^2\,\text{dm}^{-6}} = \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

The order of the units in the final answer doesn't matter: mol^{-2} dm^6 s^{-1} is equally correct.

Overall order	Example rate equation	Units of k
0	rate = k	mol dm^{-3} s^{-1}
1	rate = k[A]	s^{-1}
2	rate = k[A][B] or $k[A]^2$	dm^3 mol^{-1} s^{-1}
3	rate = $k[A][B]^2$	dm^6 mol^{-2} s^{-1}

Orders from graphs and half-lives

One way to find the order is a **continuous** experiment: follow the concentration of one reactant all the way through a single run. For this to show the order with respect to that reactant alone, every other reactant must be in large excess, so that its concentration hardly changes and has no effect on how the rate changes.

DEFINITION

The **half-life**, $t_{½}$, of a reaction is the time taken for the concentration of a reactant to fall to half its value.

Order	Concentration–time graph	Successive half-lives	Rate–concentration graph
0	straight line sloping down: the rate stays constant until the reactant runs out	each one is half the one before	horizontal line

1	curve that gets less steep (exponential decay)	**constant**, whatever the starting concentration	straight line through the origin
2	steeper at first, then a long, slow tail	each one is double the one before	upward curve through the origin (a straight line if you plot rate against $[A]^2$)

Why is the first-order half-life constant? With rate = k[A], the reaction always uses up the same *fraction* of whatever A is left in a given time. Going from 0.80 to 0.40 mol dm^{-3} takes exactly as long as going from 0.40 to 0.20 mol dm^{-3}: there is half as much to remove, but it is removed at half the rate. Radioactive decay behaves in the same way.

Using a constant half-life

1. A first-order reaction has a half-life of 25 s. The initial concentration of the reactant is 0.64 mol dm^{-3}. What is its concentration after 100 s?
2. The half-life is constant, so count half-lives: 100 s ÷ 25 s = 4 half-lives.
3. Halve the concentration four times:

$$0.64 \rightarrow 0.32 \rightarrow 0.16 \rightarrow 0.080 \rightarrow 0.040 \text{ mol dm}^{-3}$$

Equivalently, $0.64 \times (\frac{1}{2})^4 = 0.64/16 = 0.040 \text{ mol dm}^{-3}$.

For a first-order reaction the half-life and the rate constant are linked by $\ln 2 = k\,t_{1/2}$, so a half-life read from a graph gives k directly. A half-life of 25 s means $k = 0.693/25 = 0.028 \text{ s}^{-1}$. The faster the reaction (the larger k), the shorter the half-life, but it never depends on the starting concentration.

The initial rates method

The other approach is a set of separate experiments, each started with different initial concentrations, measuring only the initial rate each time (from the gradient of a tangent at t = 0, or from a clock reaction). Change **one** concentration at a time and keep the others the same, so any change in rate is caused by that reactant alone.

Experiment	[NO] / mol dm^{-3}	$[H_2]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	5.0×10^{-3}	2.0×10^{-3}	1.25×10^{-5}
2	1.0×10^{-2}	2.0×10^{-3}	5.0×10^{-5}

3	1.0×10^{-2}	4.0×10^{-3}	1.0×10^{-4}

Rate equation for $2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$

1. **Order with respect to NO:** compare experiments 1 and 2, where only [NO] changes. [NO] doubles and the rate goes from 1.25×10^{-5} to 5.0×10^{-5}, a factor of $4 = 2^2$.

   So the reaction is second order with respect to NO.

2. **Order with respect to H_2:** compare experiments 2 and 3, where only $[H_2]$ changes. $[H_2]$ doubles and the rate doubles.

   So the reaction is first order with respect to H_2.

3. Write the rate equation. The overall order is 2 + 1 = 3.

$$\text{rate} = k[\text{NO}]^2[\text{H}_2]$$

4. Find k by substituting the values from any one experiment, here experiment 1:

$$k = \frac{\text{rate}}{[\text{NO}]^2[\text{H}_2]} = \frac{1.25 \times 10^{-5}}{(5.0 \times 10^{-3})^2 \times 2.0 \times 10^{-3}} = 250\ \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

   Check with another experiment: $250 \times (1.0 \times 10^{-2})^2 \times 4.0 \times 10^{-3} = 1.0 \times 10^{-4}$, *matching experiment 3.*

When two concentrations change at once

Sometimes no pair of experiments changes only one concentration. The effects then **multiply**. Use an order you already know to strip out its effect, and whatever factor is left belongs to the other reactant.

Experiment	[P] / mol dm^{-3}	[Q] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.020	0.030	4.0×10^{-4}
2	0.040	0.030	8.0×10^{-4}
3	0.060	0.090	1.08×10^{-2}

Untangling two changes at once

1. Experiments 1 and 2: only [P] changes. It doubles and the rate doubles, so the reaction is **first order in P**.

2. Experiments 1 and 3: both change. The rate rises by a factor of $1.08 \times 10^{-2} / 4.0 \times 10^{-4} = 27$.

3. [P] has tripled (0.020 → 0.060). Being first order, that alone triples the rate. The rest of the increase must come from Q: $27 \div 3 = 9$.

4. [Q] has also tripled (0.030 → 0.090) and multiplied the rate by 9 = 3^2, so the reaction is **second order in Q**.

$$\text{rate} = k[\text{P}][\text{Q}]^2, \qquad k = \frac{4.0 \times 10^{-4}}{0.020 \times (0.030)^2} = 22 \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

Rate equations and mechanisms

Most reactions do not happen in a single collision. They go through a sequence of simple steps called the **mechanism**, often forming **intermediates**: species made in one step and used up in a later one. Adding the steps together, and cancelling the intermediates, must give the overall equation.

DEFINITION

The **rate-determining step** is the slowest step in the mechanism. The overall reaction cannot go faster than this step, just as a queue cannot move faster than its slowest checkout.

- A species that appears in the rate equation takes part in the rate-determining step, or in a step before it that makes an intermediate used in the rate-determining step.
- Its order tells you how many particles of it are involved in reaching and completing the rate-determining step.
- A reactant that is **zero order** reacts only **after** the rate-determining step. So if any reactant is zero order, the mechanism must have at least two steps.
- An intermediate never appears in the rate equation. Replace it with the reactants that formed it.
- A catalyst can appear in the rate equation, even though it is not in the overall equation, if it takes part in the rate-determining step.

Which step is rate-determining?

1. For $2NO(g) + Br_2(g) \rightarrow 2NOBr(g)$ the rate equation is rate = $k[NO]^2[Br_2]$. A two-step mechanism is proposed:

   Step 1: $NO + Br_2 \rightarrow NOBr_2$
   Step 2: $NOBr_2 + NO \rightarrow 2NOBr$

2. Check the steps add up: $NO + Br_2 + NOBr_2 + NO \rightarrow NOBr_2 + 2NOBr$. Cancelling $NOBr_2$ leaves $2NO + Br_2 \rightarrow 2NOBr$, so $NOBr_2$ is an intermediate.

3. If step 1 were the slow step, it would contain one NO and one Br_2, giving rate = $k[NO][Br_2]$. That doesn't match the experiment.

4. If step 2 is slow, it uses one $NOBr_2$ (made from one NO and one Br_2) and one more NO: two NO and one Br_2 in total. That gives rate = $k[NO]^2[Br_2]$, matching the experiment.

So step 2 is the rate-determining step, and step 1 must be fast.

Overall reaction and rate equation	Mechanism consistent with it
$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$ rate = $k[NO_2]^2$	$2NO_2 \rightarrow NO_3 + NO$ (slow) $NO_3 + CO \rightarrow NO_2 + CO_2$ (fast) CO is zero order because it only reacts after the slow step.
$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ rate = $k[H_2O_2][I^-]$	$H_2O_2 + I^- \rightarrow IO^- + H_2O$ (slow) $IO^- + H^+ \rightarrow HIO$ (fast) $HIO + H^+ + I^- \rightarrow I_2 + H_2O$ (fast)
$(CH_3)_3CBr + OH^- \rightarrow (CH_3)_3COH + Br^-$ rate = $k[(CH_3)_3CBr]$	$(CH_3)_3CBr \rightarrow (CH_3)_3C^+ + Br^-$ (slow) $(CH_3)_3C^+ + OH^- \rightarrow (CH_3)_3COH$ (fast): S_N1
$CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$ rate = $k[CH_3CH_2Br][OH^-]$	a single step in which OH^- attacks as Br^- leaves: S_N2

KEY IDEA

Kinetics is how chemists tell S_N1 from S_N2. If the rate of hydrolysis depends on $[OH^-]$, the hydroxide ion must be in the rate-determining step, so the mechanism is the one-step S_N2 route (primary

halogenoalkanes). If changing [OH^-] has no effect, the slow step is the halogenoalkane ionising on its own: S_N1 (tertiary halogenoalkanes). See the halogenoalkanes page for the mechanisms.

Core practicals: following rates in the lab

CORE PRACTICAL

The iodine–propanone reaction, followed by titration

Propanone reacts with iodine in acid solution; the acid is a catalyst:

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

Mix propanone and sulfuric acid, both in large excess, then add iodine solution and start the clock. At regular intervals withdraw a 10.0 cm^3 sample with a pipette and run it into excess sodium hydrogencarbonate solution. This **quenches** the sample by neutralising the acid catalyst. Titrate the iodine left in the sample with sodium thiosulfate solution, adding starch near the end point (blue-black to colourless):

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

- The titre is proportional to [I_2]. A graph of titre against time is a **straight line**, so the rate is constant while iodine is used up: the reaction is **zero order with respect to iodine**.
- Repeating with different concentrations of propanone or acid changes the gradient. These experiments show it is first order in propanone and first order in H^+: rate = $k[CH_3COCH_3][H^+]$.
- So the rate-determining step involves one propanone molecule and one H^+ ion, and iodine reacts in a later, fast step. H^+ appears in the rate equation even though it is a catalyst.
- The same reaction can be followed by colorimetry, since iodine is the only coloured species.
- **Safety:** propanone is highly flammable; iodine solution stains and irritates; wear eye protection.

CORE PRACTICAL

A clock reaction: initial rates and activation energy

Hydrogen peroxide oxidises iodide ions in acid: $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$. A small, fixed amount of sodium thiosulfate and some starch are added. Each iodine molecule formed is turned straight back into iodide by the thiosulfate, until the thiosulfate is used up. Then iodine builds up and the starch suddenly turns blue-black. Record the time, t.

- Every run makes the same small amount of iodine before the colour appears, so the average rate up to that point is proportional to **1/t**.
- The thiosulfate is kept small so that only a tiny fraction of the reactants is used before the colour appears. Their concentrations have barely changed, so 1/t is a good measure of the **initial** rate.
- To find an order, change the concentration of one reactant at a time (making up the volume with water so the total volume is constant) and plot 1/t against concentration.

- To find the activation energy, keep every concentration the same and repeat at several temperatures in a water bath, letting the separate solutions reach the bath temperature before mixing. Since k ∝ 1/t, a graph of ln(1/t) against 1/T has gradient $-E_a/R$ (see below).

Temperature and the Arrhenius equation

Raising the temperature increases the rate constant. The main reason is the shape of the Maxwell–Boltzmann distribution: at a higher temperature a much larger fraction of collisions have at least the activation energy, E_a. (The molecules also collide a little more often, but that is a far smaller effect.) The Arrhenius equation describes this exactly:

The Arrhenius equation and its logarithmic form

$$k = A\,e^{-E_a/RT} \qquad \ln k = \ln A - \frac{E_a}{R}\cdot\frac{1}{T}$$

k: rate constant A: Arrhenius constant (same units as k) E_a: activation energy / J mol^{-1}
R = 8.31 J K^{-1} mol^{-1} T: temperature / K

The factor $e^{-E_a/RT}$ is the fraction of collisions with enough energy to react. It is tiny when E_a is large or T is small, and it grows very rapidly with temperature. For a typical activation energy of about 50 kJ mol^{-1}, warming from 298 K to 308 K roughly doubles the rate constant.

COMMON MISTAKE

E_a is usually quoted in kJ mol^{-1}, but R is in J K^{-1} mol^{-1}. Multiply E_a by 1000 before substituting, and use T in kelvin.

Finding E_a from a graph

Compare the logarithmic form with the equation of a straight line, y = mx + c. Plotting **ln k** (y) against **1/T** (x) gives a straight line with **gradient $-E_a/R$** and intercept ln A. The gradient is always negative, and the steeper it is, the larger the activation energy. Because the rate is proportional to k at fixed concentrations, ln(rate) or ln(1/t) from a clock reaction can be plotted instead of ln k: only the intercept changes, not the gradient.

T / K	t / s	1/T / 10^{-3} K^{-1}	ln(1/t)
293	240	3.413	−5.48
303	115	3.300	−4.74

313	58.2	3.195	−4.06
323	30.6	3.096	−3.42
333	16.7	3.003	−2.82

Activation energy from clock-reaction times

1. Calculate 1/T and ln(1/t) for each run (table above) and plot ln(1/t) against 1/T. The points lie on a straight line. Draw the line of best fit.

2. Find the gradient using two points on the line that are far apart, here the first and last:

$$\text{gradient} = \frac{-2.82 - (-5.48)}{(3.003 - 3.413) \times 10^{-3}} = \frac{2.66}{-0.410 \times 10^{-3}} = -6490 \text{ K}$$

3. The gradient equals $-E_a/R$, so:

$$E_a = -\text{gradient} \times R = 6490 \times 8.31 = 53\,900 \text{ J mol}^{-1} = 53.9 \text{ kJ mol}^{-1}$$

The gradient has units of K because ln(1/t) has no units and 1/T is in K^{-1}. Multiplying K by J K^{-1} mol^{-1} gives J mol^{-1}.

Calculating the Arrhenius constant A

1. A first-order reaction has k = 2.5 × 10^{-4} s^{-1} at 320 K and E_a = 85.0 kJ mol^{-1}. Calculate A.

2. Work out the exponent, with E_a in J mol^{-1}:

$$\frac{E_a}{RT} = \frac{85\,000}{8.31 \times 320} = 31.96 \qquad e^{-31.96} = 1.31 \times 10^{-14}$$

3. Rearrange $k = A\,e^{-E_a/RT}$:

$$A = \frac{k}{e^{-E_a/RT}} = \frac{2.5 \times 10^{-4}}{1.31 \times 10^{-14}} = 1.9 \times 10^{10} \text{ s}^{-1}$$

A has the same units as k, here s^{-1}.

How much faster does a catalyst make a reaction?

1. A catalyst lowers the activation energy of a reaction from 75 kJ mol^{-1} to 50 kJ mol^{-1}. Assuming A is unchanged, by what factor does k increase at 298 K?

2. Divide the two Arrhenius expressions. A cancels:

$$\frac{k_{\text{cat}}}{k} = \frac{e^{-50\,000/RT}}{e^{-75\,000/RT}} = e^{25\,000/(8.31\times298)} = e^{10.1} \approx 2.4 \times 10^{4}$$

Lowering E_a by a third makes the reaction about 24 000 times faster, which is why catalysts matter so much in industry.

Catalysts

A catalyst speeds up a reaction without being used up, by providing an alternative route (mechanism) with a lower activation energy. It speeds up the forward and reverse reactions equally, so it does not change the position of equilibrium or the enthalpy change.

- **Homogeneous catalyst:** in the same phase as the reactants, usually all in solution. It reacts to form an intermediate and is regenerated in a later step. Because it takes part in the mechanism it can appear in the rate equation: H^+ in the iodination of propanone is an example.
- **Heterogeneous catalyst:** in a different phase from the reactants, usually a solid with gaseous or dissolved reactants. The reaction happens on its surface.

How a heterogeneous catalyst works

Reactant molecules **adsorb** (form weak bonds) onto **active sites** on the catalyst surface. This weakens bonds within the reactants, holds them close together in a favourable orientation, and concentrates them on the surface, so they react more easily. The products then **desorb**, freeing the sites for more reactant. A large surface area, from a fine powder or a thin coating on a support, gives more active sites.

The number of active sites is limited. At high pressure or concentration every site can be occupied, and adding more reactant then cannot speed the reaction up. The rate is then independent of that reactant's concentration: the reaction is zero order with respect to it. The decomposition of ammonia on a hot tungsten surface behaves like this.

Quick checks

QUICK CHECK

The rate equation for a reaction is rate = $k[X]^2[Y]$. What are the units of k?

- $mol^2\ dm^{-6}\ s^{-1}$
- $dm^3\ mol^{-1}\ s^{-1}$
- $dm^6\ mol^{-2}\ s^{-1}$
- $mol\ dm^{-3}\ s^{-1}$

The overall order is 3. k = rate/($[X]^2[Y]$) = (mol dm^{-3} s^{-1})/(mol $dm^{-3})^3$ = mol^{-2} dm^6 s^{-1}. The second option is for a second-order reaction.

QUICK CHECK

A reaction is first order with respect to A and second order with respect to B. Both concentrations are doubled at the same time. By what factor does the rate increase?

- 3
- 4
- 6
- 8

Doubling [A] multiplies the rate by 2; doubling [B] multiplies it by 2^2 = 4. The effects multiply: 2 × 4 = 8. Adding the orders (to get 3) or adding the factors (to get 6) are the usual slips.

QUICK CHECK

A first-order reaction has a half-life of 1400 s. Starting from 1.0 mol dm^{-3}, what is the concentration after 4200 s?

- 0.33 mol dm^{-3}
- 0.125 mol dm^{-3}
- 0.25 mol dm^{-3}
- 0.50 mol dm^{-3}

4200 s is three half-lives, because a first-order half-life is constant: 1.0 → 0.50 → 0.25 → 0.125 mol dm^{-3}. Dividing by 3 instead of halving three times gives the wrong answer of 0.33.

QUICK CHECK

The iodination of propanone has rate = k[CH_3COCH_3][H^+]. Which statement must be true?

- Iodine is a catalyst for the reaction.
- Iodine reacts in a step after the rate-determining step.
- The reaction has a single-step mechanism.
- Doubling [I_2] halves the rate.

Iodine is a reactant, but it is zero order, so it is not involved in the slow step and must react afterwards. A zero-order reactant means there are at least two steps. Changing [I_2] has no effect on the rate.

QUICK CHECK

A graph of ln k against 1/T for a reaction has a gradient of −7500 K. What is the activation energy? (R = 8.31 J K^{-1} mol^{-1})

- [x] 62.3 kJ mol^{-1}
- [ ] −62.3 kJ mol^{-1}
- [ ] 902 kJ mol^{-1}
- [ ] 0.90 kJ mol^{-1}

Gradient = $-E_a/R$, so E_a = 7500 × 8.31 = 62 300 J mol^{-1} = 62.3 kJ mol^{-1}. Activation energy is always positive; dividing by R instead of multiplying gives 0.90.

QUICK CHECK

In a clock reaction, why can 1/t be used as a measure of the initial rate?

- [ ] Because the reaction is zero order, so its rate never changes.
- [ ] Because the thiosulfate is a catalyst that keeps the rate constant.
- [x] Because the same small amount of product forms in every run, before the reactant concentrations have changed much.
- [ ] Because the colour appears when half of the reactants have been used up.

Rate = amount formed ÷ time. The amount (fixed by the thiosulfate) is the same in every run, so the rate is proportional to 1/t. Keeping that amount small means the concentrations are still close to their starting values, so it is an initial rate.

QUICK CHECK

Increasing the pressure of a gas reacting on a solid catalyst has no effect on the rate. What is the best explanation?

- The catalyst has been poisoned.
- The activation energy has risen to compensate.
- Gases do not adsorb onto solid surfaces.
- All the active sites on the catalyst surface are already occupied.

Reaction only happens at active sites. Once they are saturated, extra gas molecules have nowhere to adsorb, so the rate stays the same and the reaction is zero order in that gas.